



CANDIDATE
NAME

--

CIVICS
GROUP

1	7	-		
---	---	---	--	--

REGISTRATION
NUMBER

--	--

H2 Biology

Paper 4 Practical

9744/04

21 August 2018

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.
Circle the details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
1 / 2 / 3
Laboratory
Bio 1 / Bio 2 / Phy 2

For Examiner's Use	
Section A	
1	
2	
3	
Total	55

Answer **all** the questions.

- 1** Yeast cells contain enzymes which catalyse the breakdown of glucose to produce carbon dioxide and water.

The carbon dioxide reacts with water and forms a weak acid.

Bromothymol blue is a pH indicator and changes colour as shown in Table 1.1.

Table 1.1

pH	colour of bromothymol blue
8	Blue
7	Green
6	Yellow

You are required to:

- immobilise yeast cells in sodium alginate beads
- find the mean surface area of the sodium alginate beads
- investigate the independent variable, changing the surface area of the beads, on rate of enzyme reaction

You are provided with:

- yeast suspension, labelled **Y**
- sodium alginate solution, labelled **S**
- calcium chloride solution, labelled **C**
- glucose solution, labelled **G**
- bromothymol blue, labelled **B**
- sodium hydroxide solution, labelled **A**

Proceed as follows.

- 1** Put 20 cm³ of **C** into a boiling tube.
- 2** Put 5 cm³ of **S** into a small beaker or container.
- 3** Collect 5 cm³ of **Y** from below the froth and put it into the same container as **S**. Mix well.
- 4** Use a 5 cm³ syringe to collect 2 cm³ of the mixture **S** and **Y**.

- 5 Suspend the 5 cm³ syringe over the boiling tube containing **C** as shown in Fig. 1.1.

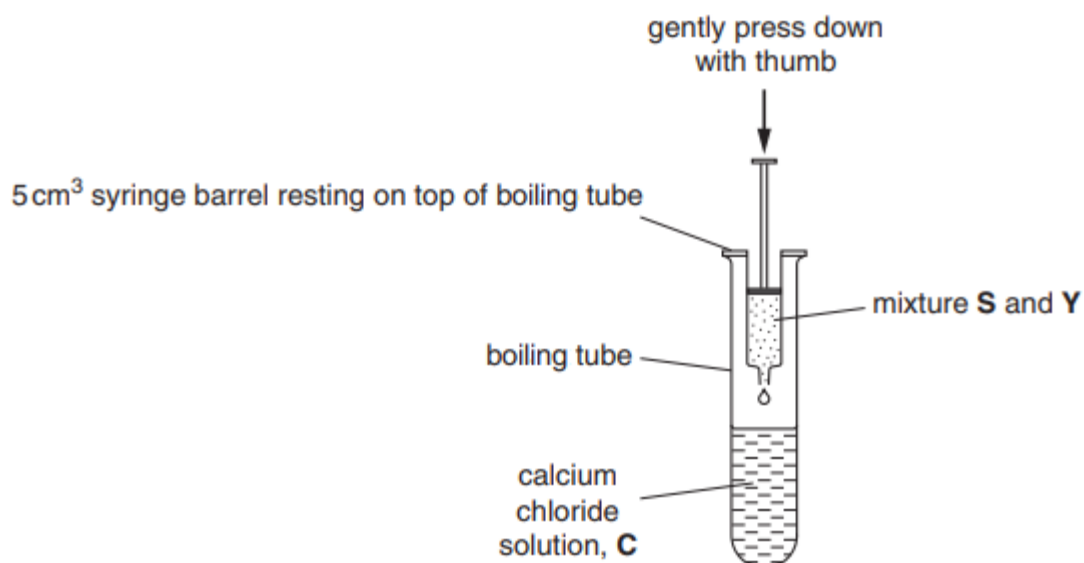


Fig. 1.1

- 6 Gently press down on the plunger of the syringe with your thumb to release a drop into solution C. The drop should form a bead.
- 7 Repeat step 6 to make the number of beads that you think you will need.
- 8 Tip the contents of the boiling tube into a Petri dish. You may use blunt forceps to pick up the beads as you select them for use in the experiment.

You will need to calculate the mean surface area of the beads.

To do this,

- decide on the number of beads you will measure
- use the 2 mm × 2 mm grid to measure each bead
- calculate the surface area of each bead using the formula
surface area = $4\pi r^2$ where $\pi = 3.14$, r = radius of a bead
- calculate the mean surface area of the beads.

In the space below, draw a bead and indicate clearly on your drawing the part of the bead you will be measuring with the 2 mm x 2 mm grid.

- (a) (i) Prepare the space below to show your measurements and calculations.

Show all the steps in your calculation of the mean.

mean surface area of the beadsmm² [4]

Before starting the investigation, read through steps 9 – 15 (including (a)(ii)) and prepare a table in (a)(iii).

A student suggested that it was possible to investigate the independent variable, surface area, by changing the number of beads. The maximum number of beads used was 20. Decide the other numbers of beads to use and state the different number of beads you will use.

(ii)
.....[1]

- 9 Label as many test-tubes as you will need with the number of beads for each test-tube.
 - 10 Put 10 cm³ of solution **G** into each test-tube.
 - 11 Put 1 cm³ of **B** into each test-tube. Attach the rubber bung to each test-tube. Mix the contents in the test tube by inverting the test tube a few times.
 - 12 If the contents of the test-tube are not blue, add one drop at a time of **A** to the contents of each test-tube to turn them all the same blue colour. Mix well after adding each drop.
 - 13 Put the required number of beads into each test-tube.
 - 14 Attach the rubber bung to each test-tube. Turn and mix the contents. Mix every 2 minutes for 6 minutes.
 - 15 Record your observations at 2 minutes interval, up to 6 minutes.
- (ii) Prepare the space below to record your observations.

[4]

(iii) The student realised that there were two independent variables in this procedure.

Other than surface area or number of beads, state the **other** independent variable.

.....
.....[1]

(iv) Suggest a suitable control that could have been used in this investigation.

.....
.....[1]

(v) Identify an error and suggest a corresponding improvement to the procedure to be carried out to improve the confidence in your results.

.....
.....
.....
.....[2]

(b) In another experiment, a students investigated the effect of varying the size of alginate beads on the rate of breakdown of hydrogen peroxide by plant catalase.

Plant extract containing catalase can be immobilised using alginate beads as well.

Other variables were considered and kept constant. There were 10 beads of each diameter.

Fig. 1.2 shows two of the sets of beads and the distribution of the plant catalase.

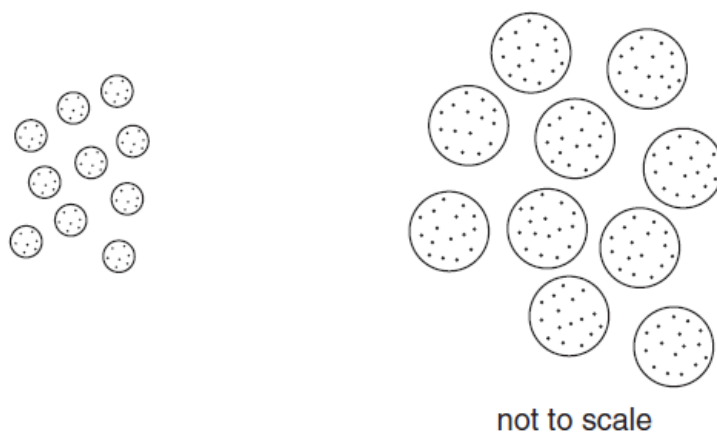


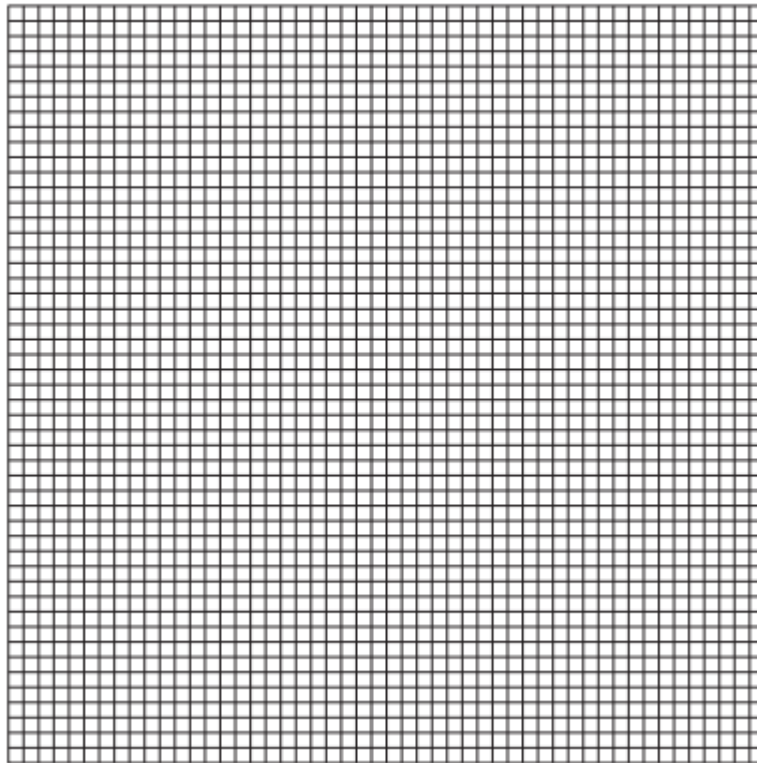
Fig. 1.2

The results of the student's investigation are shown in Table 1.2.

Table 1.2

bead diameter /mm	mass of product in a minute /mg min ⁻¹
1.0	24.0
1.5	30.5
3.0	42.0
4.5	50.0
6.0	52.5

- (i) Plot a graph of the data shown in Table 1.2.



[4]

- (ii) Using the data and Fig. 1.2, Explain the effect of immobilized plant catalase bead diameter on the mass of product formed in a minute.

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 21]

- 2 (a) Two single-celled algal species, *Scenedesmus subspicatus* and *Chlorella vulgaris* are common in polluted waters and can be used in sewage lagoons to reduce the nitrate content of the water.

A student wanted to determine the minimum amount of light that needs to be provided to each of the two types of algae for maximum growth to occur.

Design an experiment to estimate the light saturation points of both algae.

In your plan, you must use:

- a growing culture of *Chlorella vulgaris*
- a growing culture of *Scenedesmus subspicatus*
- a range of different neutral density filters
(Note: these filters each allow different percentages of light to pass through, e.g. a 10% neutral density filter allows 10% of light to be transmitted)
- a lamp that does not emit heat
- bromothymol blue indicator

You may select from the following sterilised apparatus and plan to use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.
- syringes
- timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include the layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific forms
- include reference to safety measures to minimise the any risks associated with the proposed experiment.

[9]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

9744/04/J2H2Prelim/2018

9744/04/J2H2Prelim/2018

9744/04/J2H2Prelim/2018

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

9744/04/J2H2Prelim/2018

- (b) Microalgae such as *Chlorella vulgaris* has many uses. They can serve as bioremediation agents, as feed for aquaculture, as food for humans and animals, in pigment production, in bio-removal of heavy metals and in agriculture. It is usually desirable to establish large populations of microalgae, especially in aquatic environments where they are often employed.

One means of increasing microalgal populations may be to inoculate them with other microorganisms, such as *Azospirillum brasilense*, which is known as plant-growth-promoting bacteria.

To assess the effect of the *Azospirillum brasilense* on the proliferation of *Chlorella vulgaris*, *Chlorella vulgaris* was grown with and without *Azospirillum brasilense*. The number of algal cells was counted, with 10 samples taken from each of the two culture systems every two days over a period of 10 days in total.

Table 2.1 shows the results of the investigation.

Table 2.1

Condition <i>Chlorella vulgaris</i> was grown in	Mean number of cells in 1 cm ³ at the start	Mean percentage increase in number of cells of <i>Chlorella vulgaris</i>				
		day 2	day 4	day 6	day 8	day 10
<i>Chlorella vulgaris</i> grown without <i>Azospirillum brasilense</i>	10035	40	75	140	175	220
<i>Chlorella vulgaris</i> grown with <i>Azospirillum brasilense</i>	10108	65	125	195	280	330

A *t*-test was carried out to see if there is a significant difference between the number of cells in the two culture systems at 10 days.

The value of **t = 2.62** was obtained.

- (i) State a null hypothesis for this test.

.....
[1]

- (ii) State the number of degrees of freedom for this *t*-test.

.....[1]

Table 2.2 shows some probability values of t .

Table 2.2

degrees of freedom	10	12	14	16	18	20	22	24	26	28	30	40	50	60
probability 0.05	2.23	2.18	2.14	2.12	2.10	2.09	2.07	2.06	2.06	2.05	2.04	2.02	2.01	2.00
probability 0.01	3.17	3.06	2.98	2.92	2.88	2.85	2.82	2.80	2.78	2.76	2.75	2.70	2.68	2.66

- (iii) Explain if the presence of *Azospirillum brasilense* in the growth medium is beneficial to the growth of *Chlorella vulgaris*.

.....

.....

.....

.....[2]

[Total: 13]

3 **S1** is a slide of a stained transverse section through a plant stem.

You are not expected to be familiar with this specimen.

You are required to use a sharp pencil for drawings.

- (a) (i)** Draw a large plan diagram of a sector of the stem **S1**, as shown by the shaded area in Fig. 3.1, to include **one complete vascular bundle**.

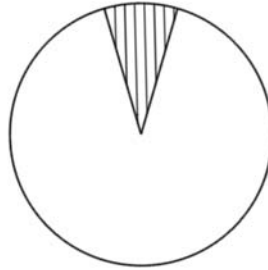


Fig. 3.1

State objective lens used:

- (ii) Use a ruler to measure your plan drawing and draw a line to show where you took the measurement. Now measure the actual size of the region where you took the measurement. Calculate the magnification of your drawing.

Show your working.

You may lose marks if you do not show clear working.

Magnification = [2]

- (b) A student working on a slide similar to S1 wanted to calibrate her eyepiece graticule for use at a particular objective lens. Fig. 3.2 shows a diagram of a stage micrometer scale that she used to calibrate the eyepiece graticule.

One division, on either the stage micrometer scale or the eyepiece graticule, is the distance between two adjacent lines.

The length of one division on this stage micrometer is **0.1 mm**.

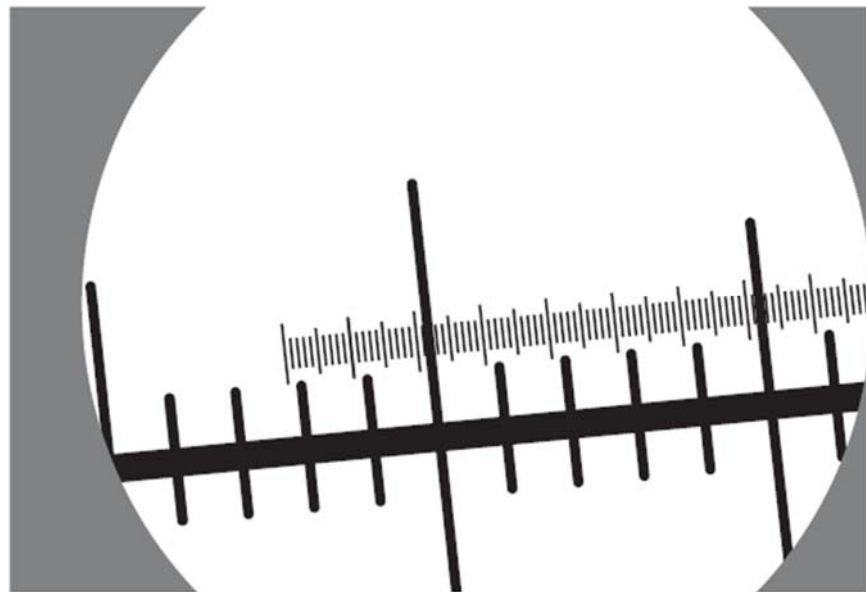


Fig. 3.2

- (i) Using Fig. 3.2, calculate the actual length of one eyepiece graticule unit by following the steps below. [2]

Step 1

Length of 1 eyepiece graticule unit = mm divided by = mm

Step 2

Convert answer from Step 1 to a unit that is more suitable for use in light microscopy:

(answer from step 1) multiplied by =

Fig. 3.3 is a photomicrograph showing a part of plant organ from another plant taken using the same objective lens from the same microscope in Fig. 3.2.

- (ii) Use the calibration of the eyepiece graticule from **(b)(i)** and **Fig. 3.3** to calculate the actual length of the plant tissue from X to Y.

You will lose marks if you do not show all steps in your calculation and if you do not use the appropriate units.

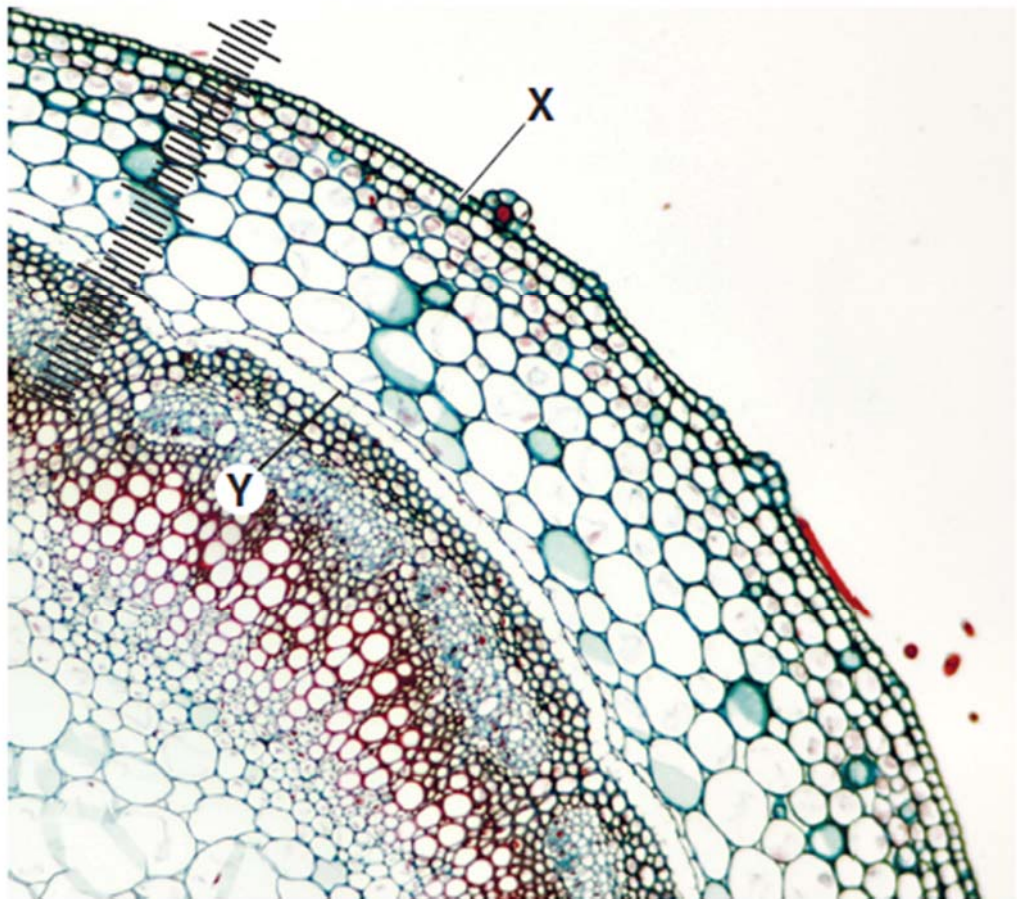


Fig. 3.3

Working:

[2]

- (c) Fig. 3.4 is a photomicrograph of a vascular bundle similar to that in slide **S1**. Using slide **S1**, locate and draw **five adjacent** circular structures in the region indicated by the label **T** in Fig. 3.4. Label each of these five structures **A** to **E**.

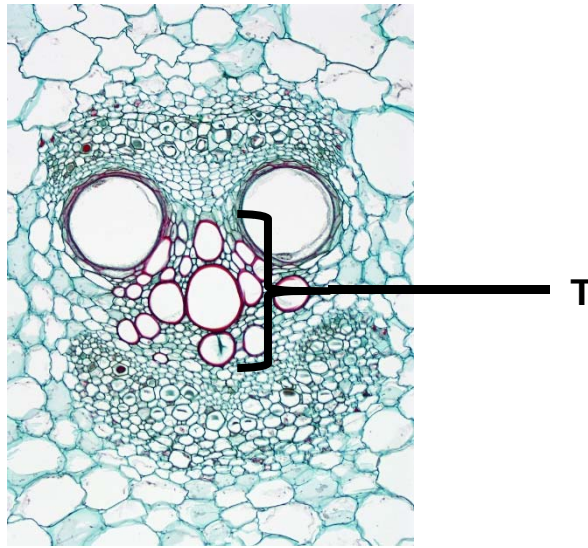


Fig. 3.4

State the objective lens used:

- (d) Use the eyepiece graticule to measure the diameter of each of the five circular structures you have drawn.

Calculate the mean diameter of the circular structures and the standard deviation (s). Calculate standard deviation using the formula:

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Σ = 'sum of' $\bar{x}$ = mean

n = sample size (number of observations)

Record your measurements and the results of your calculations in the space below.

You may lose marks if you do not show clear working.

Mean diameter of circular structures =

Standard deviation of circular structures =

[2]

- (e) Fig. 3.5 is a photomicrograph showing a stem from another plant.
You are not expected to be familiar with this specimen.

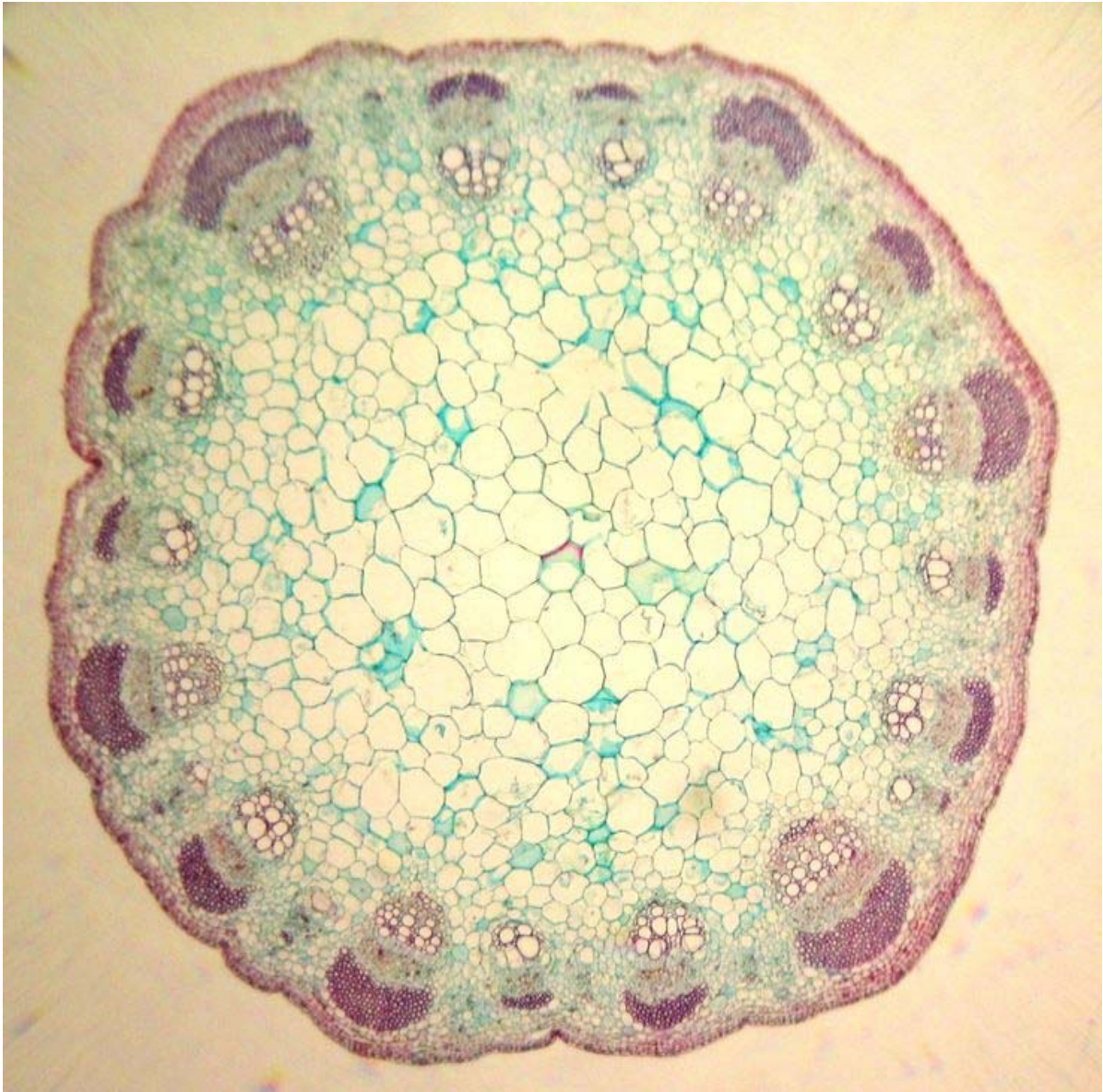


Fig. 3.5

Prepare the space below so that it is suitable for you to compare **three** observable differences between **S1** and **Fig. 3.5**.

- End of Paper -